

Advances In MOTORS AND DRIVERS For Wearables



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Ever wondered how your mobile vibrates? Let us see. There is a small motor in your phone, which is responsible for shaking it every time you receive a notification. I observed this phenomenon in a Motorola phone by removing the back cover and calling that number. Interestingly, there is a half-cut cylinder (cut along its axis) attached to the motor's axle that rotates very fast to generate the vibration.

Motion is the main motive behind using motors. In embedded systems an area that has seen extensive use of motors is robots. If you go through the schematics of a humanoid robot, you will probably forget the motor count by the time you are through with the review. Every joint is essentially a motor, with some having more than one.

Types and advances

Let us take the example of the humanoid robot. We would require a multitude of motors for different use cases. For in-

stance, a type of motor would be used for lifting heavy parts of a machine, whereas another for controlled motions of its individual parts. These would then, in turn, require similarly different drivers for their application.

A trend that is catching up in system design using motors is employment of direct current (DC) motors. "Ranging from automobiles and fans to wearables, we are seeing an increase of brushless DC (BLDC) motors," says Hrishikesh Kamat, chief executive officer, Shalaka Technologies.

With their increased employment in embedded system designs, the size of motors has become smaller than ever through continuous iterations and technology advances. Motors and their drivers are moving towards smaller, smarter, safer and faster solutions. "Drivers are coming inbuilt with Hall sensor inputs and back electromagnetic force sensing capability," says T. Anand, managing director, Kneutron Technologies. He adds, "This allows designers to reduce part counts and, thus, BOM costs."

Drivers can also be integrated with radio frequency chipsets for remote control in home automation and industrial applications, enabling remote diagnostics of motor drives as well.

Sensor-less control

Kamat says, "Implementation of sensor-less control of BLDC motors is seen in many of the recent applications. Elimination of rotor positioning sensors like Hall sensors not only reduces cost but also increases reliability." Sensor-less drive is based on detection of back electromagnetic force induced by the movement of a permanent magnet rotor in front of stator winding.

InstaSPIN is another technology that monitors the motor's flux to determine

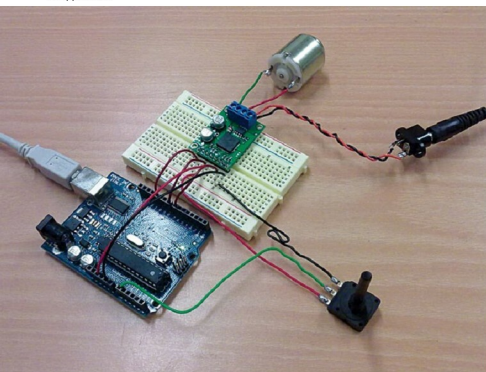


Fig. 1: Motor drivers in application